LESSON 09 · MISSION GUIDE

The Living Ledger

pandas: the whole quest ledger in one variable, with filters, sorting and Excel in and out.

From **Pell, Quartermistress of the Guild**
To **Jen, Keeper of the Guild Ledger**
By **raven, regarding the fourteen tabs**

Keeper, you've seen the old ledger workbook. Fourteen tabs. "Quests", "Quests (2)", "Quests OLD DO NOT USE", "Pell working", "Dalvric working" (locked, password lost at sea), and one Biscuit renamed by sitting on the keyboard. Every moon I filter, sort, add my helper columns, copy to a new tab, and hope the Royal Auditors don't ask which tab is real.

Orrin says pandas gives you a **DataFrame**: the whole ledger in one variable. Filter it, sort it, add columns, read it from Excel, write it back out, never click "Enable editing". A ledger that lives, he says. I said that sounds like something Dalvric would say. He went quiet. The workbook is `data/guild_ledger.xlsx`. Don't rename the sheets; the Grimoire still reads them. — P.

What you'll be able to do after today

- [ ] Read a CSV or an Excel sheet into a **DataFrame** and look at it: `.shape`, `.head()`, `.describe()`
- [ ] Pick a **column** (`df["reward"]`) or several (`df[["a", "b"]]`) and summarise it
- [ ] **Filter rows** the AutoFilter way, with masks, `.isin()` and `.str.contains()`
- [ ] **Add columns** computed from other columns, no dragging
- [ ] **Sort** by one or more columns, and find the top or bottom N
- [ ] Write a DataFrame to CSV, and turn a text column into real **dates**

Why this matters at work

- **This replaces the helper tabs.** Filter, sort, add columns: the things you do by hand every month are one line each, and the script does them identically next month.
- **Excel in, Excel out.** pandas reads the real workbook and writes files Excel opens. Nobody else has to change anything.

Four big ideas

1. DataFrame = sheet, Series = column	`df` is the table. `df["reward"]` is one column, a Series, which is a NumPy array with a name and an index.
2. The index is the row label	The numbers down the left: start at 0, stick to their rows when you filter or sort. Not Excel row numbers.
3. Masks, exactly like NumPy	`df[df["reward"] > 1000]`. Combine with `&` and `|`, brackets round each comparison.
4. Most things return a new DataFrame	`df.sort_values(...)` gives you a sorted copy; `df` itself is unchanged. Chain them: `df[mask].sort_values("x").head()`.

Three things that will bite you

- Column names are **case-sensitive** and must match exactly: `df["Region"]` is a KeyError if the column is `region`.
- `df["a", "b"]` is wrong; `df[["a", "b"]]` (a list inside the brackets) is right.
- `.to_csv(...)` without `index=False` writes the index as a column. You'll get "Unnamed: 0" back.

How today works

15 MIN
Lecture
Lars opens the ledger and does the Quartermistress's month in five lines

60 MIN
Exercise sheet
Seven ravens about the 400 quests, the heroes and the old workbook

15 MIN
Pages from the Grimoire, page 9
Dalvric's last ever script, note: "should work"

The ravens

#	FROM	RAVEN	YOU'LL USE
1	Pell	Open the ledger	`read_csv .shape .columns .head() .tail()`
2	Pell	One column at a time	Series, `.mean() .nunique() .describe()`
3	Siv	AutoFilter	masks, `.isin() .str.contains()`
4	Pell	New columns	`df["new"] = ...`
5	The Queen	Sorting	`.sort_values()`, `ascending=`
6	Orrin	Excel in, CSV out	`read_excel .to_csv`
7 ⭐	Pell	Real dates	`pd.to_datetime .dt.month .dt.day_name()`

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if Biscuit's theory can wait), each printing the expected output
- [ ] Dalvric's Ironhills report runs and prints the expected five lines, with one commit per bug
- [ ] Everything is pushed to `lesson-09` and you've opened a pull request



Next lesson: The War Table. `groupby` is the PivotTable, `merge` is XLOOKUP, and together they give you success by monster, crowns by region, loot by hero. The Queen will want all three.